BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 246 · Computational problem recognition

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why may Spiral be preferable for an uncertain high-risk project?

Q2. What is abstraction?

Q3. What is a base case?

Q4. For C# array int[] x={2,4,6,8}, a loop doubles elements at even indexes (0,2). Give the final array and sum.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

problem recognition — Determine what problem is being solved and its computational form.

decomposition — Split into manageable subproblems.

divide and conquer — Divide input, solve smaller instances, combine results.

constraint — Condition a valid solution must satisfy.

tractability — Whether an algorithm can solve realistic sizes with feasible resources.

New learning

Recognise ambiguity — 'Fair' is subjective until objective criteria are supplied, such as balancing first-period lessons across teachers.

Make requirements testable — Represent rooms, lessons and slots; specify hard constraints and a success test.

Select techniques appropriately — Decompose into assign/check/evaluate modules. Divide and conquer is useful where subproblems can be solved and combined consistently; not every decomposition is divide and conquer.

Worked example

Rewrite 'choose the best delivery van route' into a computational task.

1. Predict the method — Replace vague 'best' with a measurable objective such as minimum total travel time.

2. Check and explain — Specify route constraints and verify a produced route covers all required locations.

We Do · Practise with guidance

A school wants the 'fairest' allocation of laptops across classes. Make it computable.

```
Inputs: stock, demand per class, priorities
Output: allocations respecting stock
Criterion: minimise unmet essential demand
```

Predict the method

Use explicit demand and stock data and define what fairness means with measurable priorities.

Check and explain

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Check total allocated never exceeds stock and each class allocation follows the stated policy.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

For a system allocating parking spaces, give one input, one constraint, one output and one test of correctness.

Question 2

Explain why 'select the nicest library books' is not computable as stated; propose precise inputs, a ranking rule and an output.

Question 3

Compare decomposition and divide-and-conquer for sorting 1,000 items, using merge sort as an example.

Exit Ticket · Answer before checking

An unclear human judgement becomes computable simply by choosing a faster processor.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It prioritises identifying and addressing risks in each cycle.

Q2. Selecting relevant detail and omitting irrelevant detail.

Q3. A condition that ends recursion.

Q4. Final [4,4,12,8]. Sum=28. Indexes 0 and 2 changed; 1 and 3 unchanged.

You Do pages · Independent questions

Q1. [4] Input available bays/vehicle requests [1]; constraint one vehicle per bay [1]; output bay assignments [1]; test no duplicate assignments [1].

Q2. [6] 'Nicest' is subjective [1]; need measurable criteria [1], such as age-suitable genre [1] and prior loan count [1]; rank by agreed score [1]; output ordered ISBN list [1].

Q3. [6] Decomposition splits the overall task into subproblems [1]; divide-and-conquer recursively splits input [1], sorts sublists [1] and merges them [1]; combined result is sorted [1], recurrence gives O(n log n) work for merge sort [1].

Exit Ticket · Model answer

False. It must first be expressed with measurable inputs, rules and success criteria.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding